EE2213 Introduction to AI

Lecture 12: Introduction to Knowledge Representation and Reasoning

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**
  - What is AI
  - Applications of AI
  - AI agent
- **Search (Shaojing)**
  - Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
  - Informed search algorithms: greedy best-first, A*
  - Applications
- **Optimisation (Shaojing)**
  - Linear programming
  - Convex problems
  - Applications
- **Machine learning (Wang Si) (Weeks 6-9)**
  - Supervised and unsupervised learning: regression, classification, clustering
  - Neural networks and deep learning
  - Applications
- **Knowledge representation (Wang Si) (Weeks 10-11)**
  - Knowledge Representation and Reasoning
  - Propositional Logic
  - Applications
- **Ethical considerations (Shaojing)**
  - Bias in AI
  - Privacy concerns
  - Societal impact

AGENDA

- We will discuss:
  - Definition of Knowledge Representation and Reasoning (KRR)
  - Types of Knowledge Representation
  - Types of Reasoning
  - A Case Study: The Wumpus World

At the end of this lecture, you should be able to:

- ✓ Describe the role of KRR in AI agent
- ✓ Differentiate among types of knowledge, types of knowledge representation, and types of reasoning
- ✓ Understand Knowledge-based Agent

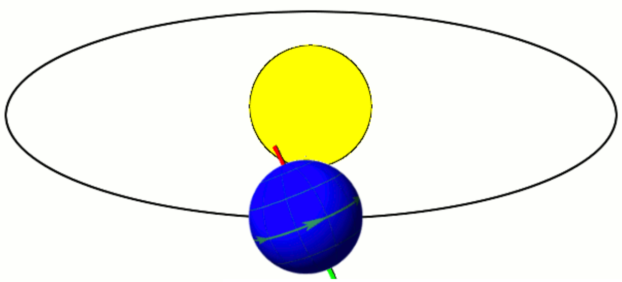
What does it mean to ‘know’ something?

Knowledge

➢ What is Knowledge?

An awareness of facts, a familiarity with individuals and situations, or a practical skill

----by Wikipedia



What if we want machines to know things?

Knowledge Representation and Reasoning (KRR)

➢ What is KRR?

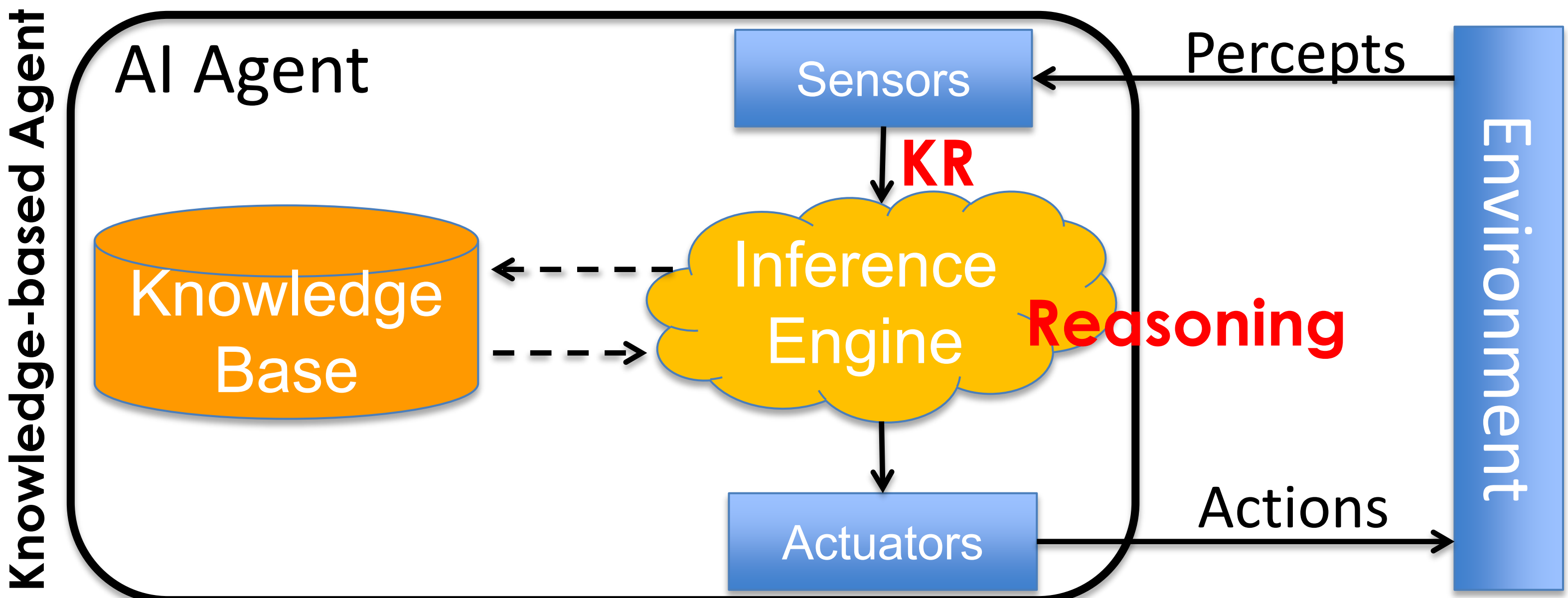
A field in AI concerned with how knowledge can be **formally** represented and used by machines to reason and make decisions

Part1: KR

How we store and organize knowledge

Part2: Reasoning

how to draw conclusions or make decisions from it

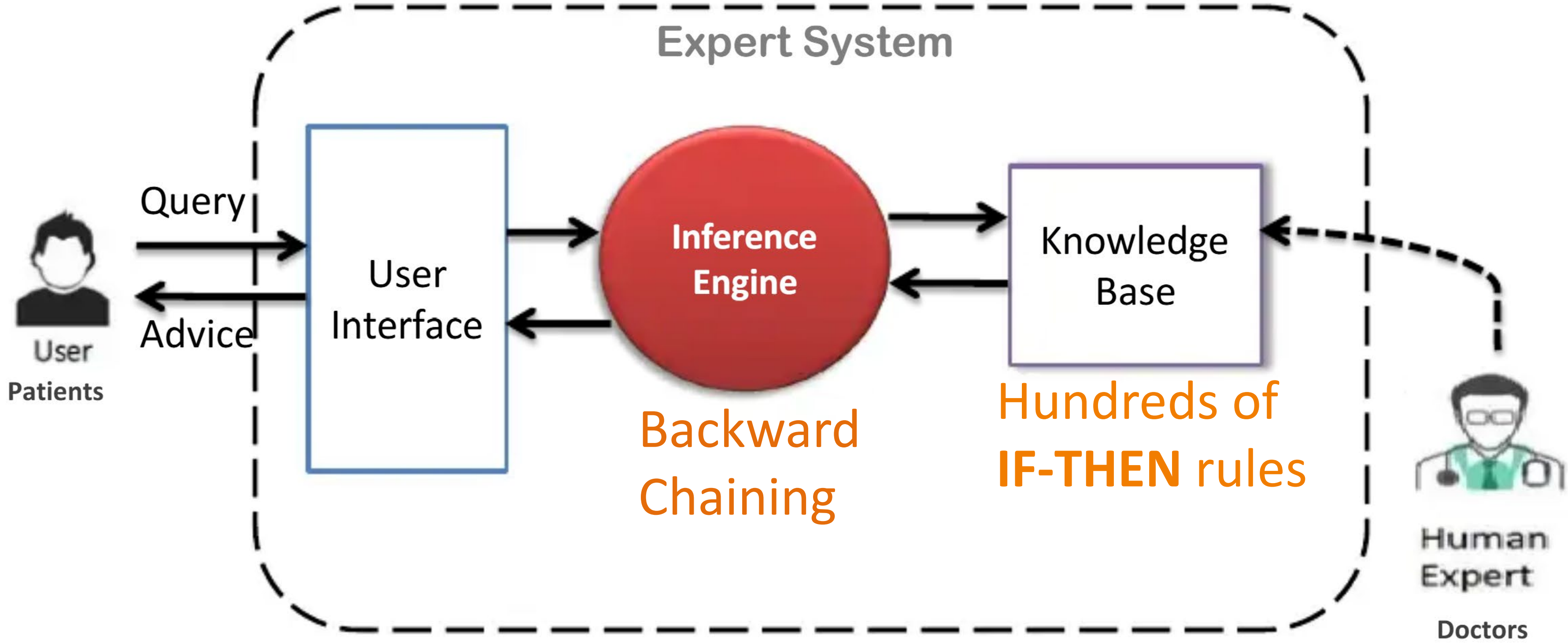


Knowledge Representation and Reasoning (KRR)

➢ KRR in AI Agent

<u>Example 1: MYCIN Expert System</u> (KR: **IF-THEN rules**)

(developed in the 1970s to diagnose bacterial infections and recommend antibiotics)

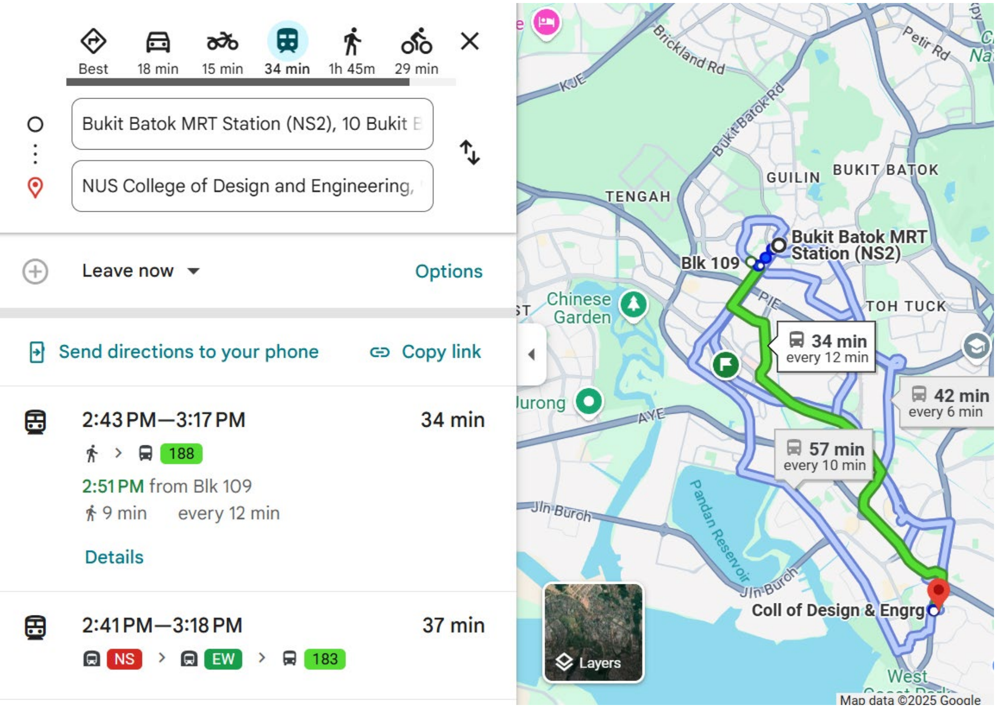


e.g., IF the patient has a fever AND high white blood cell count, *THEN* there is a high chance of bacterial infection.

Knowledge Representation and Reasoning (KRR)

➢ KRR in AI Agent

Example 2: Google Maps' Route Recommendation



KR:

- Road Network as a **Graph**
  - ✓ nodes = intersections;
  - ✓ edges = roads;
  - ✓ Edge attributes = distance, speed limit, average traffic, road type, toll info, etc.
- Historical and Real-time traffic Data
  - ✓ Real-time traffic info as **dynamic weights on edges.**
  - ✓ Historical traffic pattern as **time-stamped edge attributes/probabilistic model/lookup tables**

⋮

Reasoning:

Graph search algorithms like Dijkstra's or A*

Dynamic adaptation for real-time traffic

Predictive reasoning to predict future traffic condition

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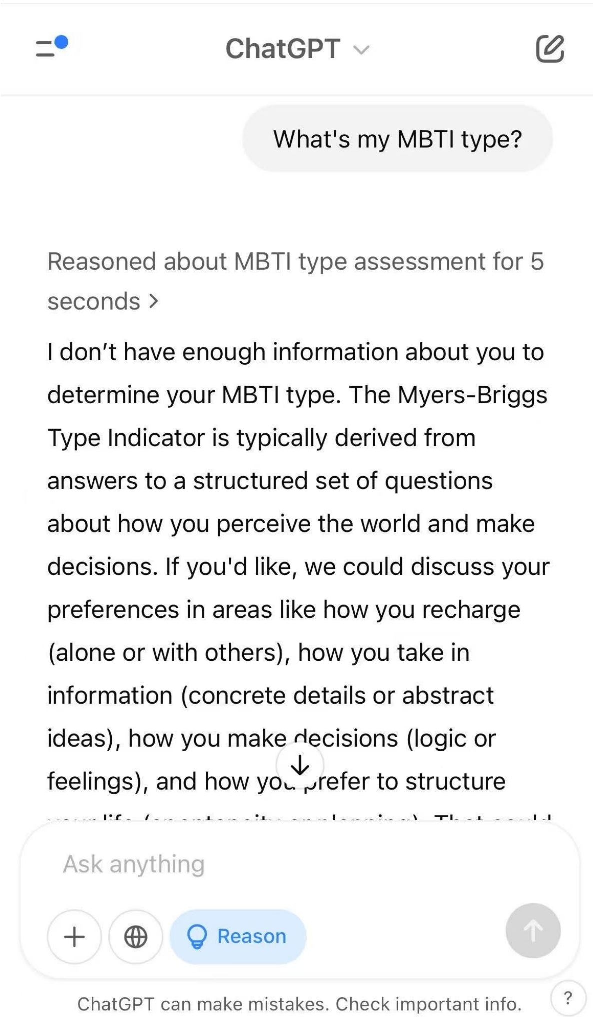
Knowledge Representation and Reasoning (KRR)

➢ KRR in AI Agent

Example 3: ChatGPT



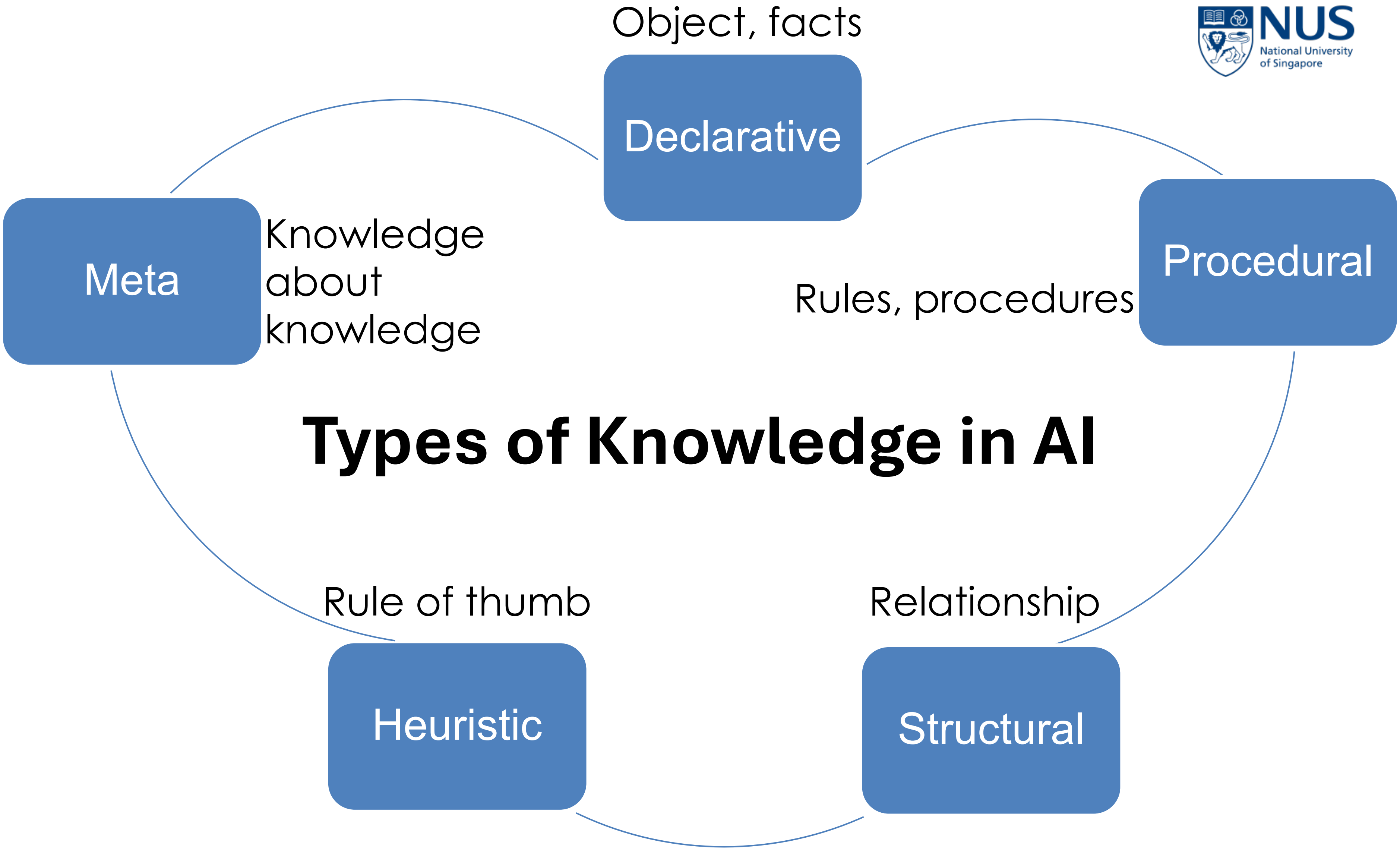
KR: Vector-based representation

(facts, concepts, and language patterns are encoded in the weights of the neural networks)

Reasoning: Predictive reasoning

"By predicting the most likely next token in a sequence, conditioned on context"

Types of Knowledge in AI



Types of Knowledge in AI

➢ Declarative Knowledge

Facts and statements about the world

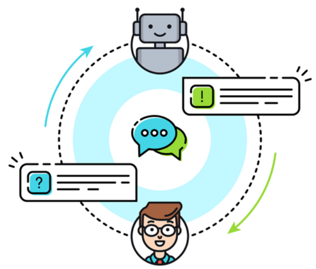
(“Knowing that” something is true)

Examples
“Beijing is the capital of China”
“Water freezes at 0°C”
"Dogs are mammals"
"A triangle has 3 sides"

Applications:

1. Form the knowledge base of knowledge-based agent (e.g., expert system)

2. Allow chatbots to answer factual questions.



3. Specify the world state and constraints for robot navigation system.

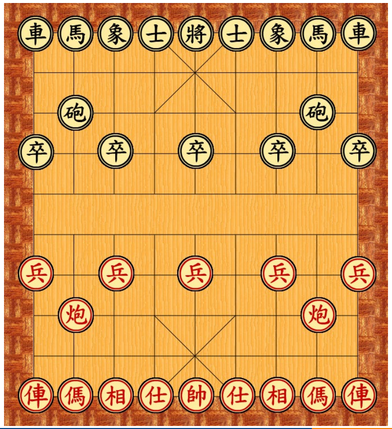
“Room A is connected to Room B via Door 1”

“A corridor leads from the entrance to the lab”

“Do not enter hazardous zones”

“Keep 1m away from walls”

4. Define rules and facts of the game for the game AI



Types of Knowledge in AI

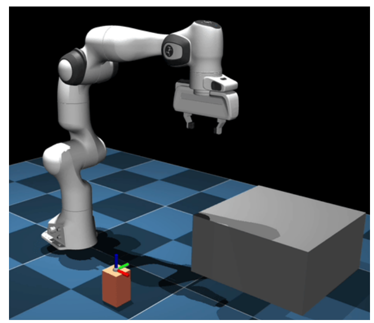
➢ Procedural Knowledge

Rules, strategies, and sequence of actions that can be directly applied to tasks.

(“Knowing how” to do something)

Examples
Steps to solve a Rubik’s Cube
How to sort a list
How to grasp a cup of water
How to navigate from A to B

Applications:



1. Motion control for robotics

2. Problem-Solving

3. Guide parsing and generation of the sentence in Natural Language Processing

4. Behavior scripts for NPCs in Game AI

5. Machine Learning

   “Steps to update weights in neural networks.”

Types of Knowledge in AI

➢ Structural Knowledge

Relationships and hierarchies between concepts.

Examples
"A robin is a bird, a bird is an animal, so a robin is an animal."
“Fuel and oxygen are required for fire; fire produces smoke; smoke can trigger alarms."
“A car is a type of vehicle, all vehicles have wheels, therefore a car has wheels."
“A day has three main meals: breakfast, lunch, and dinner; breakfast comes before lunch, and lunch comes before dinner.”

Applications:

1. Recommender System

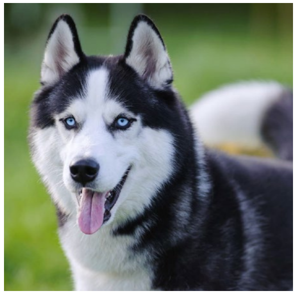
   “Action movies → Superhero → Marvel”

2. Allow chatbots to answer indirect questions

   “Alice is Bob’s mother.” “Bob is Charlie’s father.”

   “Who is Charlie’s grandmother?”

3. Hierarchical Classification



Types of Knowledge in AI

➢ Heuristic Knowledge

Rules of thumb or experience-based shortcuts
(not guaranteed to be correct/optimal, but useful/fast)

Examples
"Explore the most promising node first in A*."
"If a door is locked, try the keys in order of most frequently used keys first."
"Choose the restaurant with the longest queue."
"Carry an umbrella if the sky looks cloudy."

Applications:

1. Recommender System
   "Recommend popular items first"

2. Robot Arm Planning
   "prefer moves that minimize distance to the target object"

3. Pathfinding

Types of Knowledge in AI

➢ Meta-Knowledge

Know how to use, organize, select, or reason with knowledge

(Knowing about knowledge)

Examples
"Use supervised learning if labeled data is available; otherwise, use unsupervised learning"
"If the GPS signal is weak, I'll follow road signs instead."
"When I am tired, I'll do easier tasks first."
"If my first approach to a puzzle isn't working, I'll try a different strategy."

Types of Knowledge in AI

- Practice Questions: Identify type of knowledge

"If it looks like a duck and quacks like a duck, assume it's a duck"

"To make coffee: add grounds, pour water, heat."

"Birds can fly"

"Penguins are birds, and birds are animals."

"Penguins are birds."

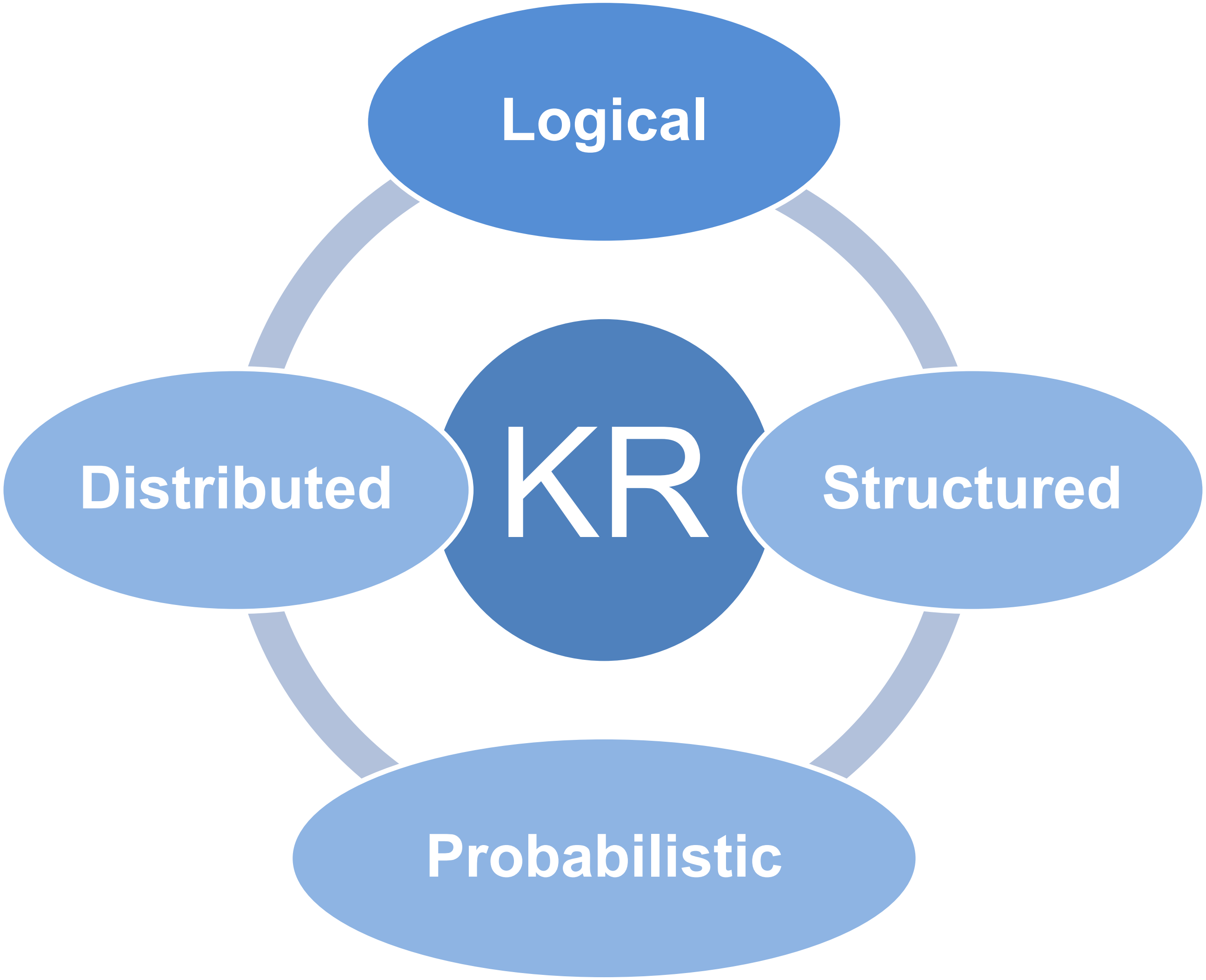
"This system works poorly when data is noisy."

Knowledge Representation

- Encode information into a format that computers can understand and use to solve problems

Knowledge Representation

➤ Core Methods



Knowledge Representation

➢ Logical KR

Represent knowledge using formal logic---clear rules or statements that are either **true** or **false**.

Techniques: Propositional logic, First-order logic

declarative statements (propositions) linked by logical operators like AND, OR, NOT, etc.

e.g., "IF it rains AND the ground is wet, THEN the road is slippery"

A: "It rains"
B: "The ground is wet"
C: " The road is slippery"

$$A \wedge B \Rightarrow C$$

Extends propositional logic by introducing variables, quantifiers, and predicates.

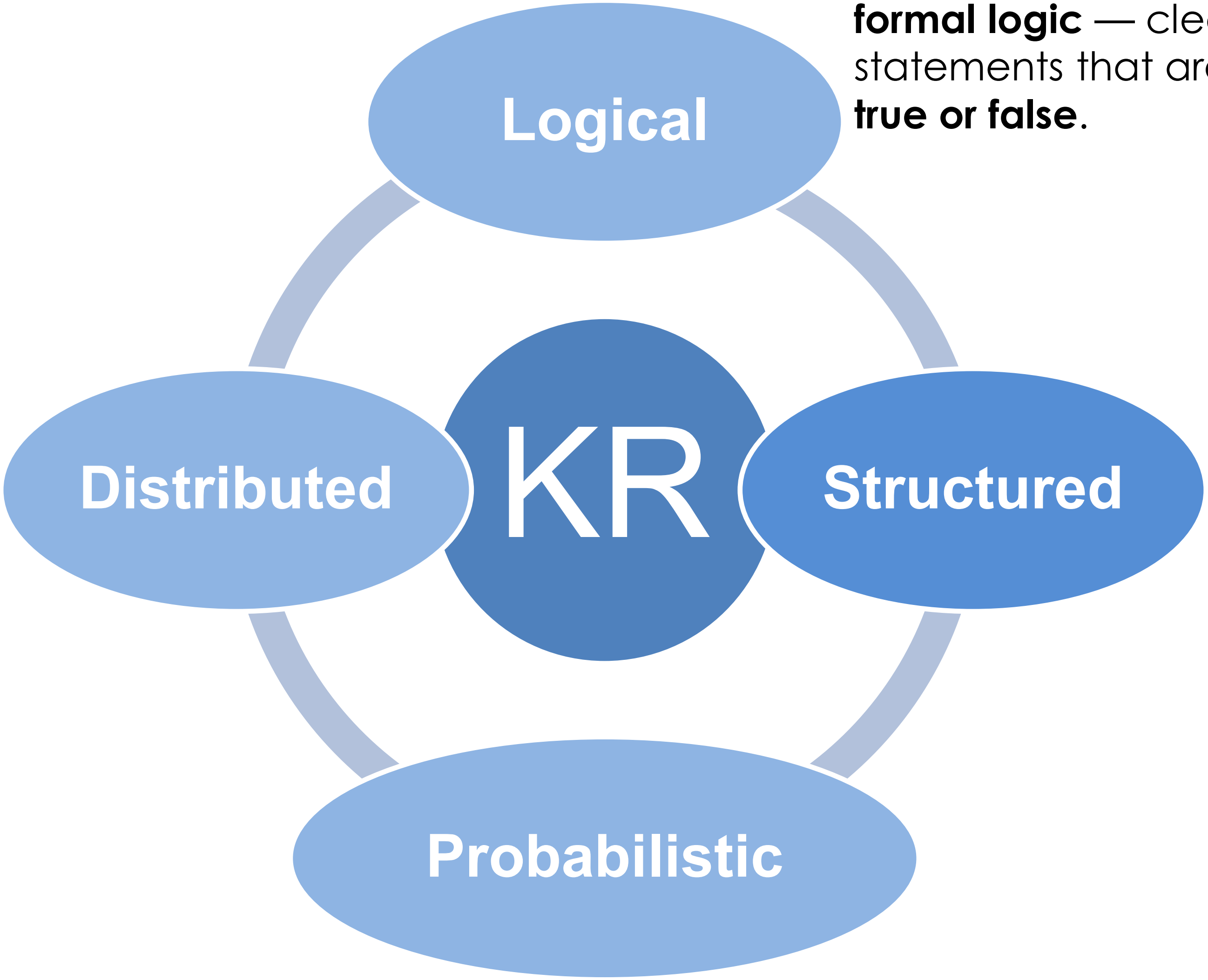
e.g., "All humans are mortal"

x: variable
$Human(x)$: x is a human
$Mortal(x)$: x is mortal

$$\forall x\ Human(x) \Rightarrow Mortal(x)$$

Knowledge Representation

➢ Core Methods



formal logic — clear rules or statements that are either **true or false**.

Knowledge Representation

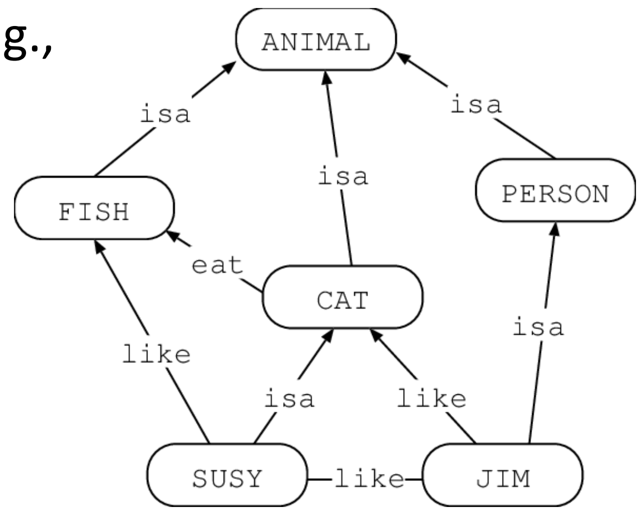
➢ Structured KR

Organize knowledge into structured formats like hierarchies, graphs, or frames.

Techniques: Semantic Networks, Frames, etc.

Nodes = Concepts
Edges = Relationships

e.g.,



Structured object with slots & values

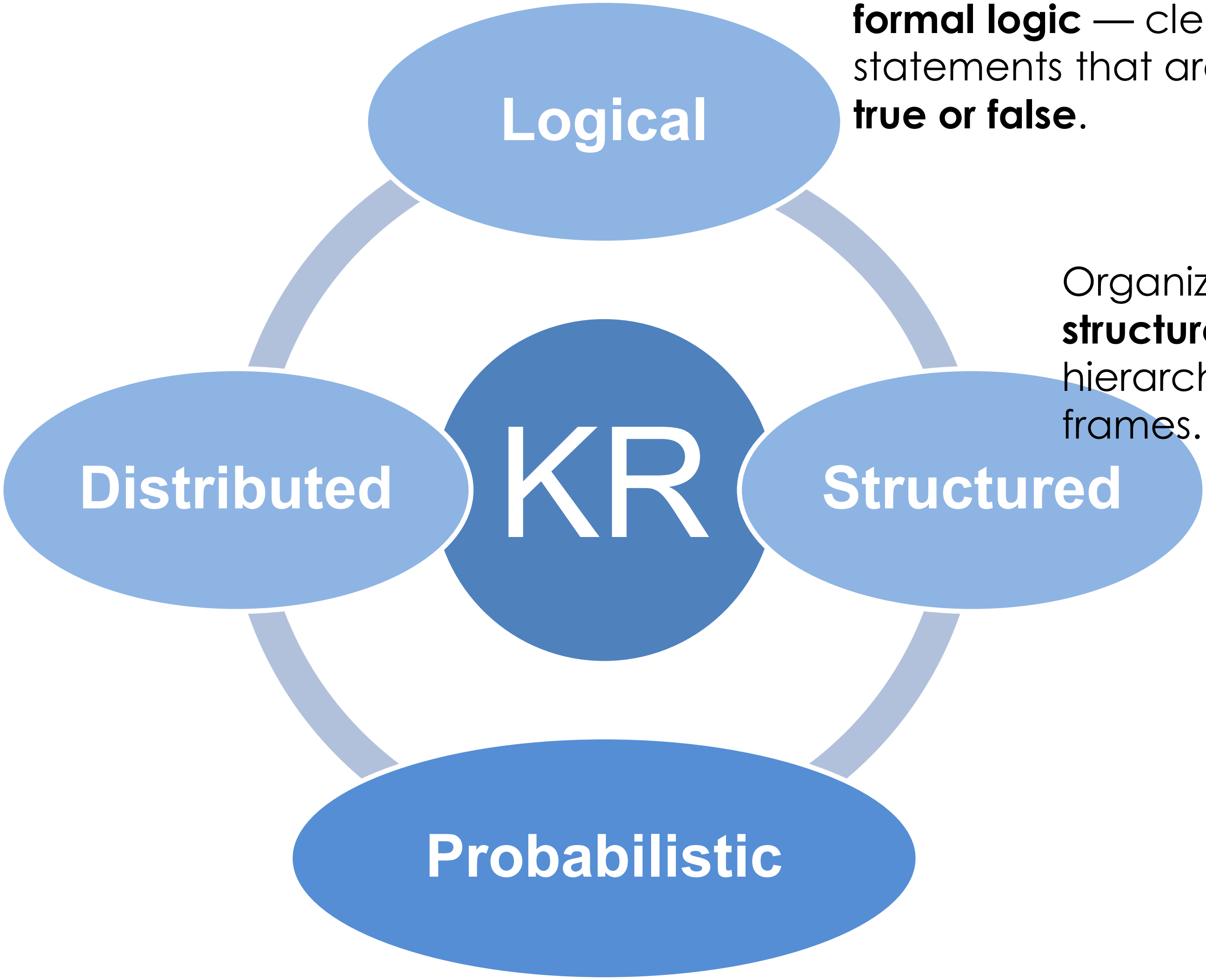
e.g.,

University	
Name	NUS
Programs	B.E, M.E, Ph.D
Department	ECE, CEE

slots values

Knowledge Representation

➢ Core Methods



formal logic — clear rules or statements that are either **true or false**.

Organize knowledge into **structured formats** like hierarchies, graphs, or frames.

Knowledge Representation

➢ Probabilistic KR

Represent knowledge where uncertainty exists by assigning probabilities to outcomes.

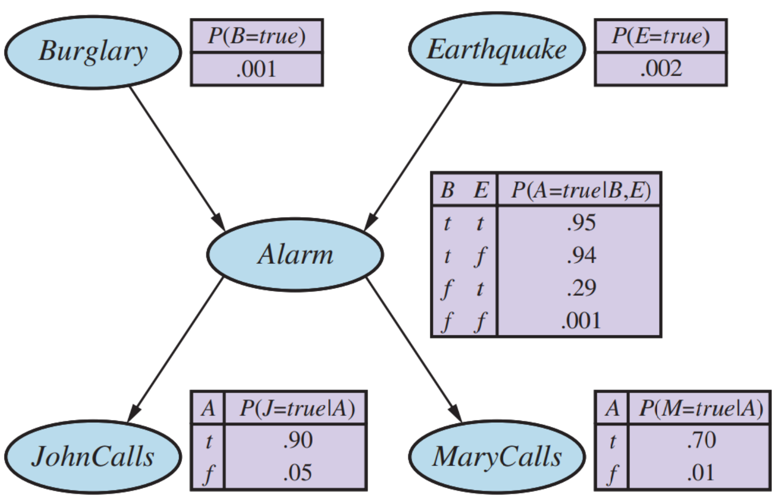
Techniques: Bayesian Networks, Markov Networks, etc.

Model causal relationships among variables (directed graph)

Nodes = Variables

Edges = Conditional dependencies

e.g.,

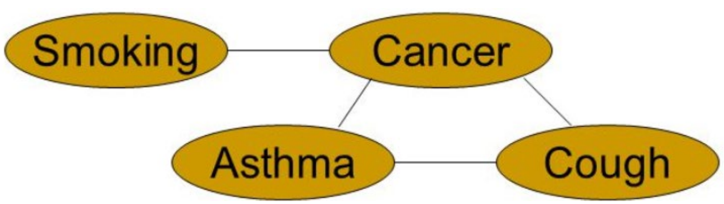


Model correlations among variables (undirected graph)

Nodes = Variables

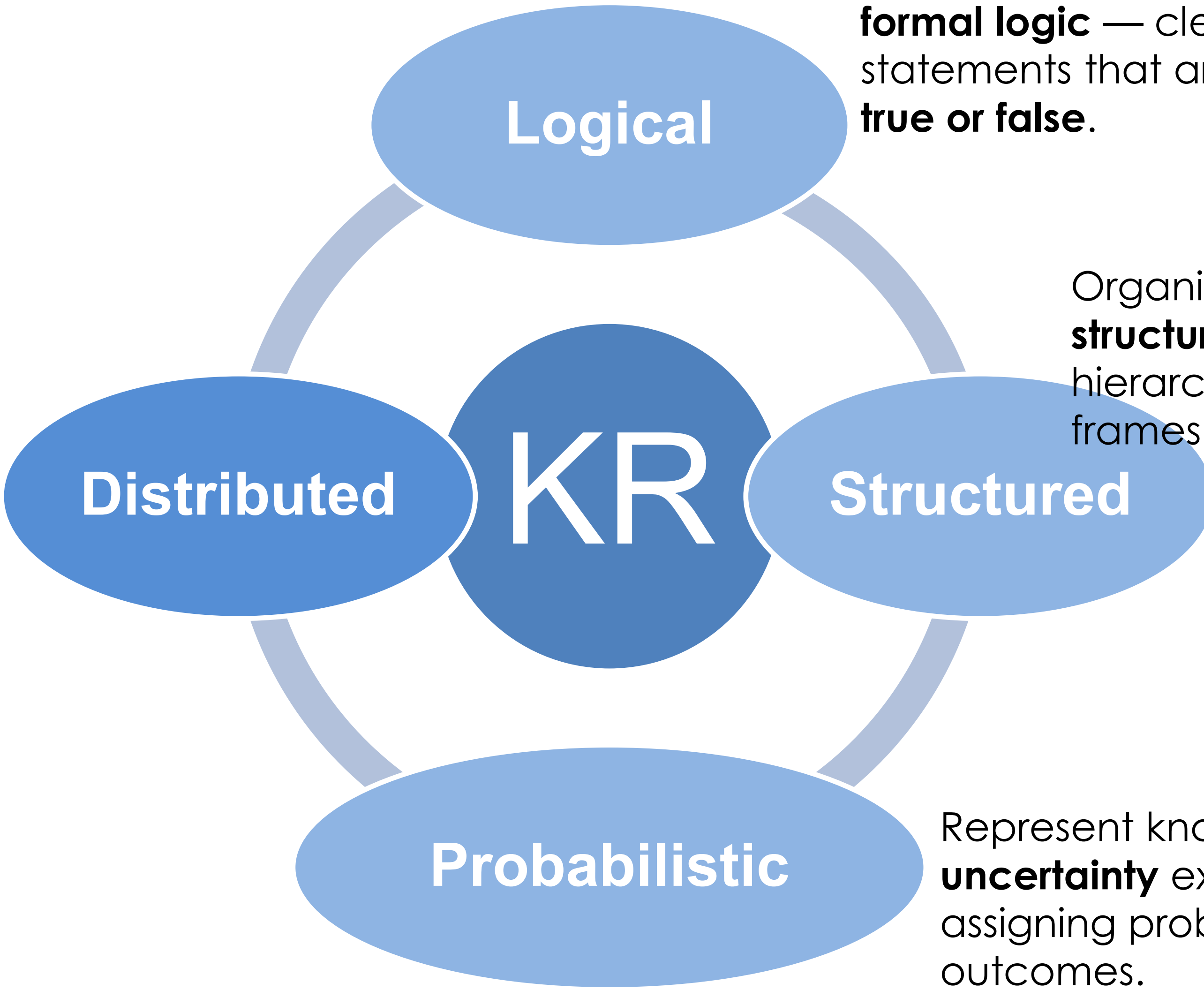
Edges = Direct probabilistic influence

e.g.,



Knowledge Representation

➢ Core Methods



formal logic — clear rules or statements that are either **true or false**.

Organize knowledge into **structured formats** like hierarchies, graphs, or frames.

Represent knowledge where **uncertainty** exists by assigning probabilities to outcomes.

Knowledge Representation

➢ Distributed KR

Represent knowledge as vectors in a high-dimensional space, often learned from data (not hand-coded)

Techniques: Word embeddings, neural networks

e.g., Word2Vec map synonyms to nearby vectors

Word	Vector (Simplified)
king	[0.52, 0.71, -0.33, 0.44, 0.91]
queen	[0.51, 0.69, -0.31, 0.46, 0.93]
man	[0.60, 0.80, -0.40, 0.38, 0.90]
woman	[0.49, 0.70, -0.29, 0.47, 0.92]

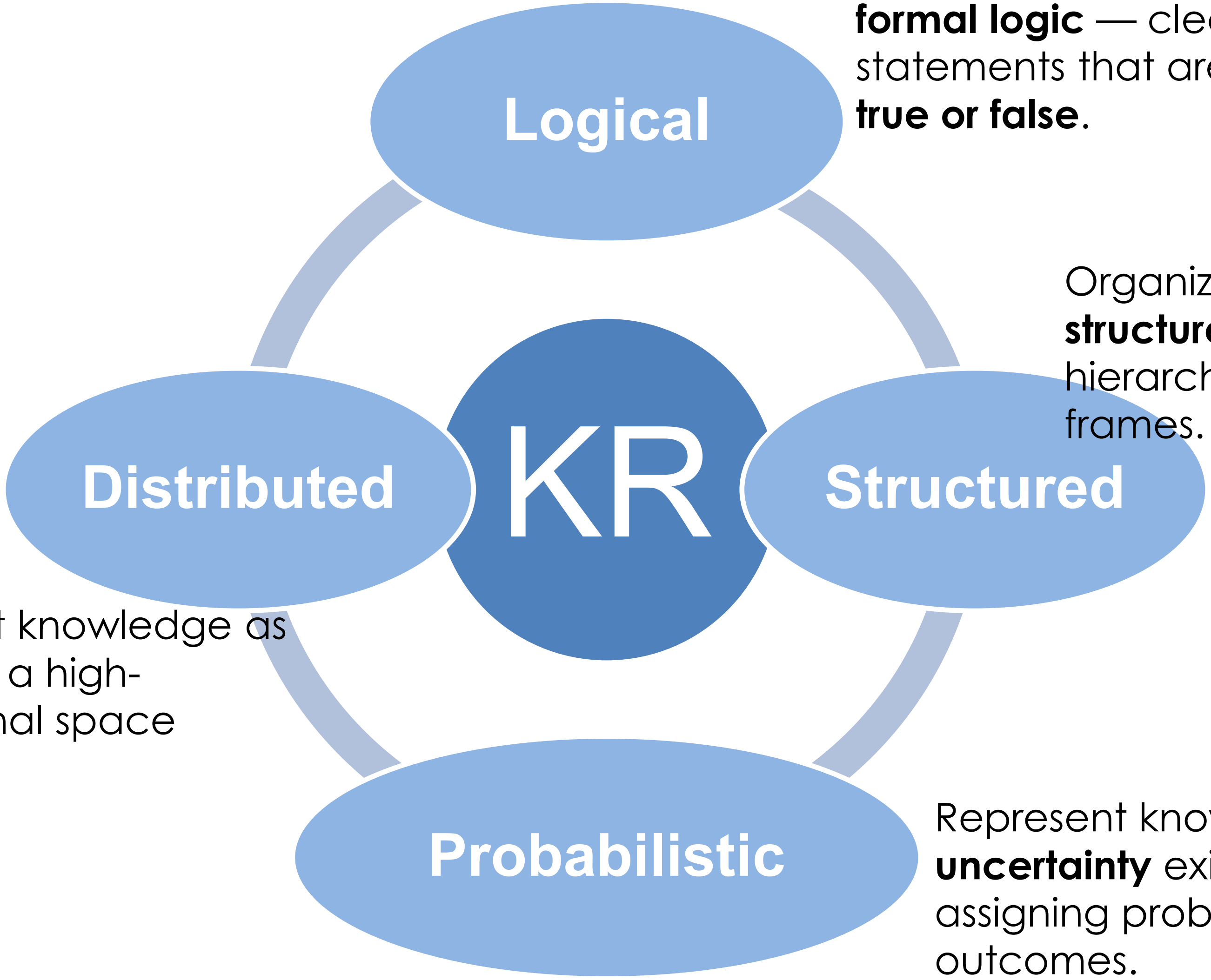
king - man + woman
≈ queen

Knowledge Representation

➢ Core Methods

formal logic — clear rules or statements that are either **true or false**.

Organize knowledge into **structured formats** like hierarchies, graphs, or frames.



Represent knowledge as **vectors** in a high-dimensional space

Represent knowledge where **uncertainty** exists by assigning probabilities to outcomes.

Knowledge Representation

- Practice Questions: Identify the type of knowledge representation

Imagine a Self-driving Car:

- It uses a neural network to detect road signs from images.
- It estimate probabilities like "There is an 85% chance that the object ahead is a pedestrian.
- It uses rules like "If the light is red, stop."
- It stores info about itself like:

Vehicle	
Manufacturer	Tesla
Model	Model 3
Sensors	Cameras, LIDAR, Radar, Ultrasonic

Reasoning

- Draw conclusions, make inference, and solve problems based on the existing knowledge

Reasoning

➢ 3 Main Types of Reasoning

Deductive

General Rule

↓

Specific Conclusion

Inductive

Specific Observation

↓

General Conclusion

Abductive

Incomplete Observation

↓

Best Explanation

e.g.,

All humans are mortal
Wang Si is a human
↓
Wang Si is mortal

Swan 1 is white;
Swan 2 is white...
↓
All swans are probably white

The grass is wet
↓
It probably rained

Reasoning

➢ 3 Main Types of Reasoning

Deductive	Inductive	Abductive
General Rule	Specific Observation	Incomplete Observation
↓	↓	↓
Specific Conclusion	General Conclusion	Best Explanation
Logical KR	Probabilistic KR Distributed KR	Probabilistic KR

Reasoning

- Practice Questions: Identify the type of reasoning

  - A student consistently scores well in the previous exams. The teacher predicts that the student will also score well in the next exam
  - A mechanic finds that a car won’t start. Possible causes: dead battery, empty fuel tank, faulty ignition. He hypothesizes it’s most likely a dead battery
  - All even numbers are divisible by 2. 8 is an even number. Therefore, 8 is divisible by 2.

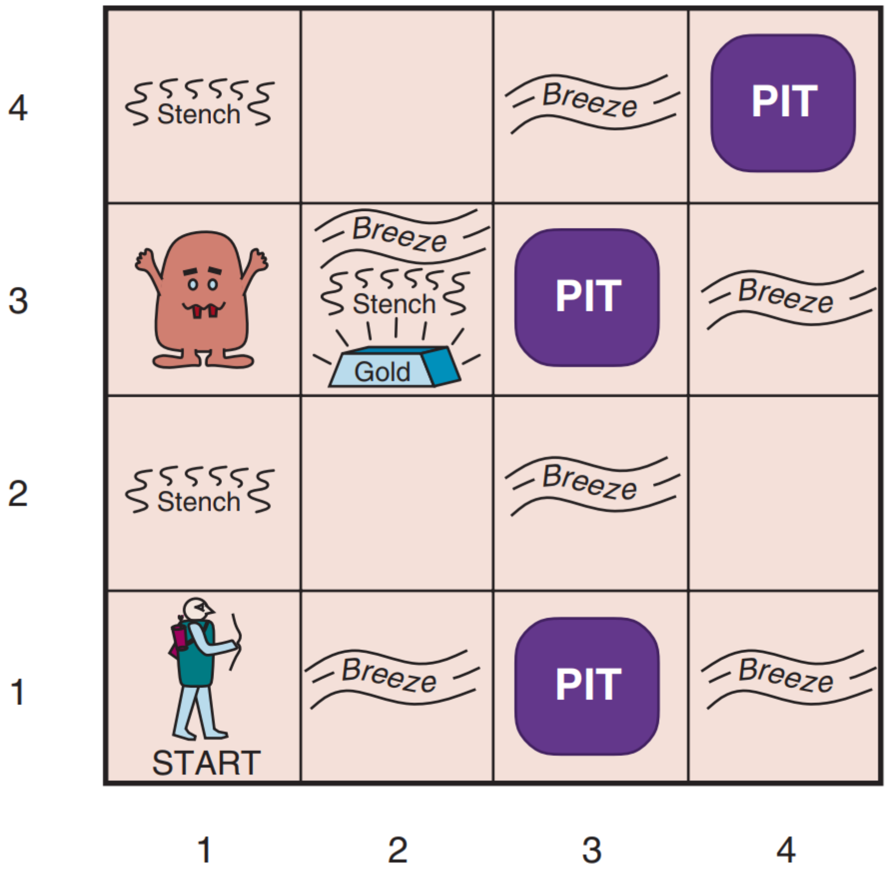
function KB-AGENT(*percept*) **returns** an *action*
  persistent: *KB*, a knowledge base
    t, a counter, initially 0, indicating time

  TELL(*KB*, MAKE-PERCEPT-SENTENCE(*percept*, *t*))
  action ← ASK(*KB*, MAKE-ACTION-QUERY(*t*))
  TELL(*KB*, MAKE-ACTION-SENTENCE(*action*, *t*))
  $t \leftarrow t + 1$
  return *action*

Knowledge-based Agent:

Case Study

The Wumpus World



Environment:

- 4x4 grid of rooms
- The agent starts in [1,1], facing rightward.
- Some rooms contain:
  - 💀 A Wumpus —kill the agent if entered.
  - 🕳 Pits — deadly if fallen into.
  - 💰 Gold — the agent wants

Actuators:

- Move, grab gold, shoot an arrow, and climb out.

Sensors:

- Stench (adjacent to Wumpus),
- Breeze (adjacent to PIT),
- Glitter (Gold),
- Bump (Wall),
- Scream (Wumpus is killed)

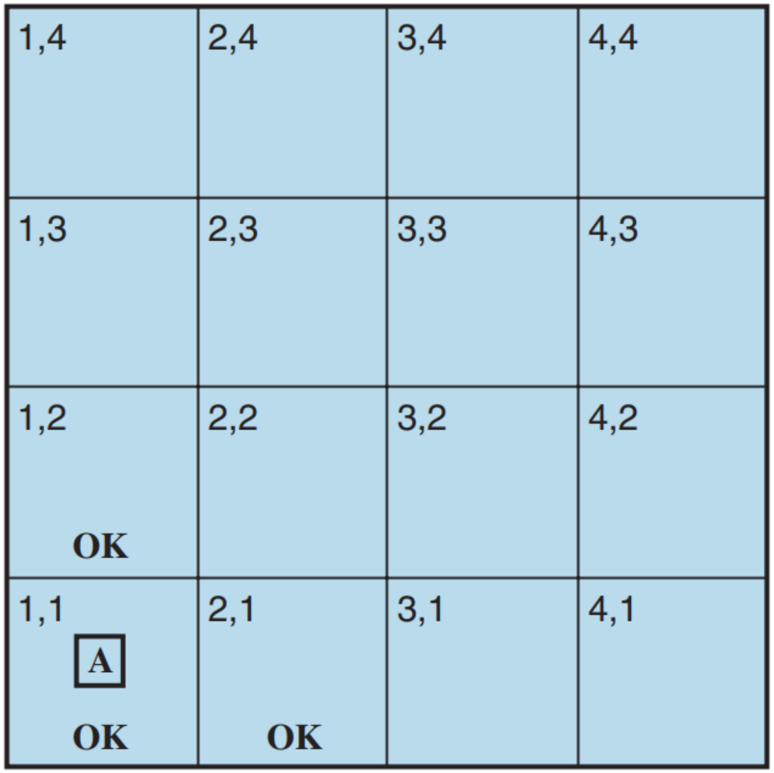
Performance Measure:

+1000: climb out with gold
-1000: dead
-1: each action taken
-10: use up the only arrow

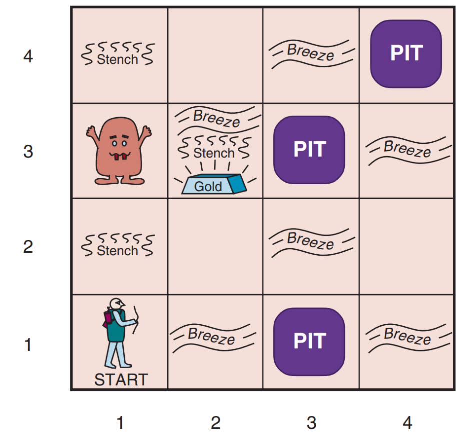
The Wumpus World

- Environment Exploration by a Knowledge-based Wumpus agent

Initial situation



A = *Agent*
B = *Breeze*
G = *Glitter, Gold*
OK = *Safe square*
P = *Pit*
S = *Stench*
V = *Visited*
W = *Wumpus*



KB: rules of the world;
the agent is in [1,1]; [1,1] is OK

Percept: [None, None, None, None, None]

Inference: [1,2] and [2,1] are OK

Action: Move to [2,1]

UPDATE

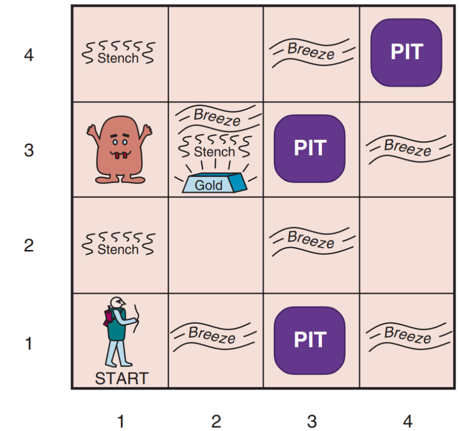
The Wumpus World

- Environment Exploration by a Knowledge-based Wumpus agent

After 1st Move

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 $\boxed{A}$ B OK	3,1 P?	4,1

$\boxed{A}$ = *Agent*
B = *Breeze*
G = *Glitter, Gold*
OK = *Safe square*
P = *Pit*
S = *Stench*
V = *Visited*
W = *Wumpus*



KB: rules of the world;
the agent is in [2,1];
[1,1], [1,2], [2,1] are OK

Percept: [None, Breeze, None, None, None]

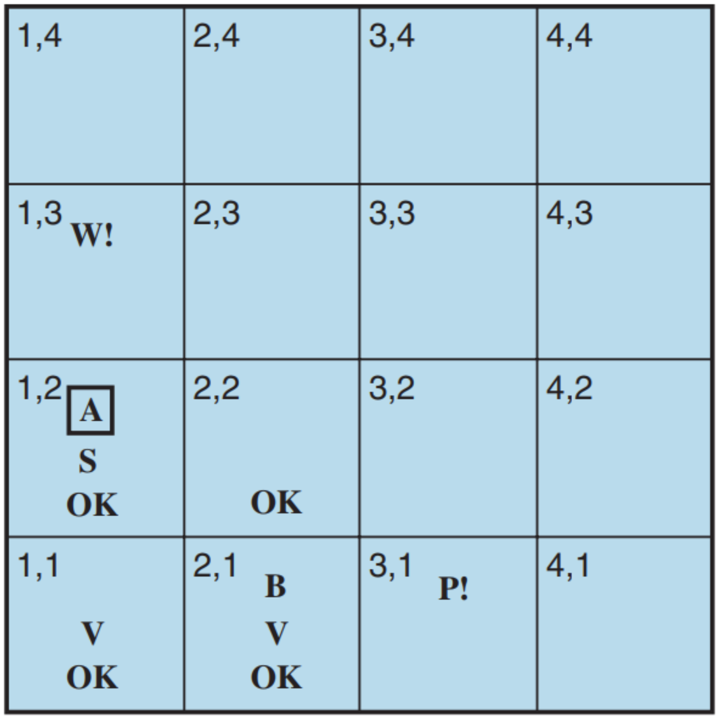
Inference: A Pit in [2,2] or [3,1] or both,
No Wumpus in [2,2] and [3,1]

Action: Move back to [1,1], and then proceed to [1, 2]

The Wumpus World

- Environment Exploration by a Knowledge-based Wumpus agent

After 3rd Move



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V = *Visited*
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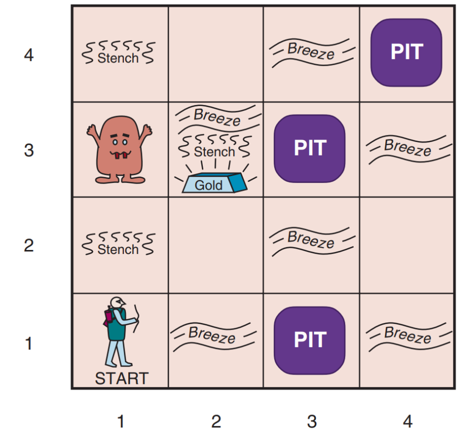

KB: rules of the world;
the agent is in [1,2];
[1,1], [1,2], [2,1] are OK;
[2,2] and [3,1] are P? But no Wumpus.

Percept: [Stench, None, None, None, None]

Action: Move to [2, 2]

Inference: Wumpus in [1,3];
[2,2] is OK;
A Pit in [3,1]

The Wumpus World

- Environment Exploration by a Knowledge-based Wumpus agent

After 4th Move

1,4	2,4	3,4	4,4
1,3 W!	2,3 OK	3,3	4,3
1,2 V S OK	2,2 $\boxed{A}$ OK	3,2 OK	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

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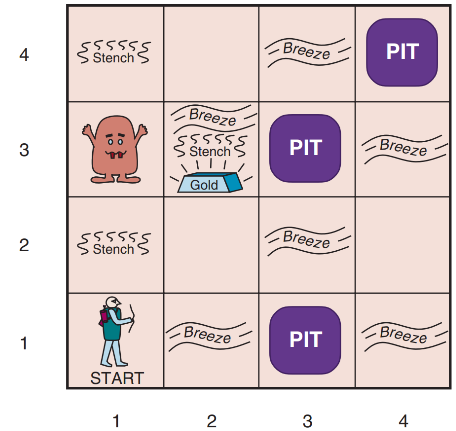
V = *Visited*
W = *Wumpus*



KB: rules of the world;
the agent is in [2,2];
[1,1], [1,2], [2,1], [2,2] are OK;
A Pit in [3,1]; A Wumpus in [1,3]

Percept: [None, None, None, None, None]

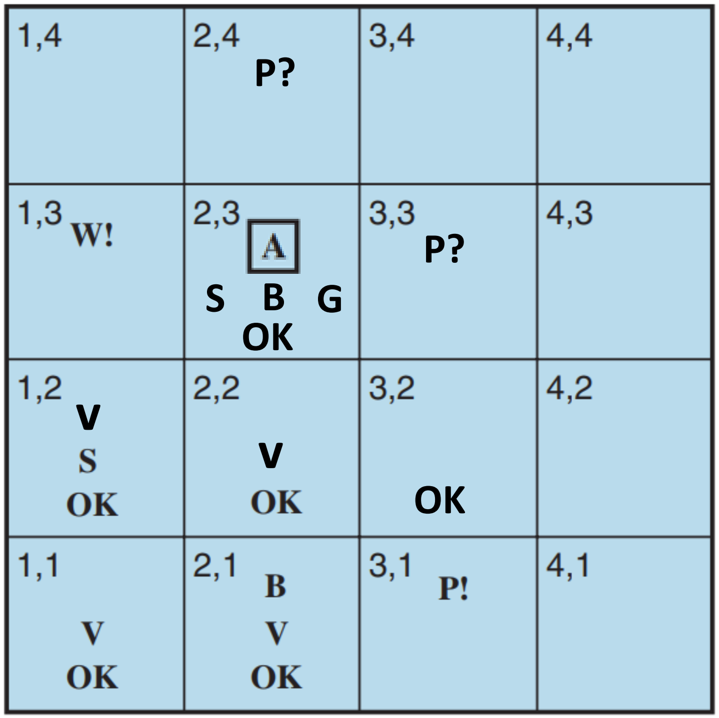
Action: Move to [2, 3]

Inference: [2,3] and [3,2] are OK.

The Wumpus World

- Environment Exploration by a Knowledge-based Wumpus agent

After 5th Move



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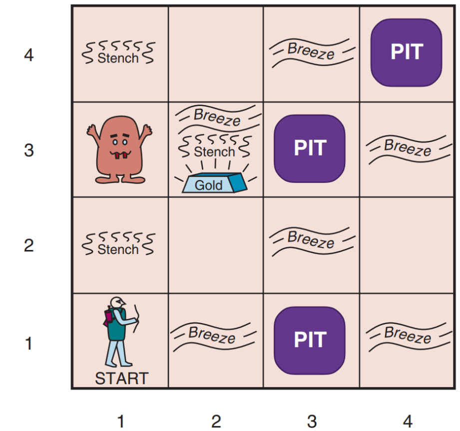

KB: rules of the world;
the agent is in [2,3];
[1,1], [1,2], [2,1], [2,2], [2,3], [3,2] are OK;
A Pit in [3,1]; A Wumpus in [1,3]

Percept: [Stench, Breeze, Glitter, None, None]

Action: Grab gold, and move back to [1, 1], and climb out

Inference: [2,3] has gold;
A Pit in [2,4] or [3,3] or both

Summary

- ✓ 5 types of Knowledge

  Declarative, Procedural, Structural, Heuristic, Meta

- ✓ 4 types of Knowledge Representation

  Logical, Structured, Probabilistic, Distributed

- ✓ 3 types of Reasoning

  Deductive, Inductive, Abductive

- ✓ 1 Case Study of Knowledge-based Agent: The Wumpus World

THE END